

# VERIQO MIP-001 Implementation Report

**Round:** Master Mandate — Final Integrated Architecture + Master Implementation Plan **Inputs:** two specifications, implemented in the order given — VERIQO\_FINAL\_INTEGRATED\_ARCHITECTURE.docx (doc 1) then VERIQO\_MASTER\_IMPLEMENTATION\_PLAN.docx (doc 2) **Agent completion report format:** MIP-001 §38 **Status taxonomy:** MIP-001 §34 — no capability reported as `DONE` without a level

## 1. IMPLEMENTED

### Doc 1 → W0 Architecture Freeze (complete)

Seven documents, the deliverable set W0 names exactly:

| Document                                    | What it fixes                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------|
| docs/architecture/VERIQO_OS_ARCHITECTURE.md | Five planes, canonical lifecycle, and a binding map from each plane onto code that already exists |
| docs/constitution/EVIDENCE_CONSTITUTION.md  | 30 articles, enforcement classes, dependency graph                                                |
| docs/ieap/IEAP-001.md                       | Acquisition protocol, fail-closed rules, three lineage dimensions                                 |
| docs/eqf/EQF-001.md                         | The epistemic frame                                                                               |
| docs/nep/NEP-001.md                         | Neutral Evidence Protocol                                                                         |
| docs/ai/AI-AUTHORITY-001.md                 | The AI authority boundary                                                                         |
| docs/mip/MIP-001.md                         | The §37 handoff artifact: current → target → gap, per wave                                        |

The reframing doc 1 asks for is now the repository's stated architecture: **VERIQO is an Evidence Operating System**, and maritime, commodity, insurance, finance, dark web and dispute are domain intelligence capabilities consuming one evidence fabric — not fifteen products sharing a logo.

### Doc 2 → MIP P0 constitutional layer (seven new packages)

| Package                         | Wave       | What it does                                              |
|---------------------------------|------------|-----------------------------------------------------------|
| pkg/constitution                | §4         | The 30 articles as <b>executable checks</b>               |
| pkg/contract/event              | §10, §11   | Canonical <b>EventEnvelope</b> + 25-family taxonomy       |
| pkg/qualification/reverseproof  | W4.1       | Claim → conditions → requirements → gap                   |
| pkg/qualification/observability | W4.2       | The nine-condition gate; observed absence                 |
| pkg/qualification/independence  | W4.3       | 15 dimensions, same-root clustering                       |
| pkg/qualification/state         | W4.5, §29  | 9-state vocabulary + single-source exception              |
| pkg/qualification/nextbest      | §21        | Priority ranking, hard filters first                      |
| pkg/disclosure/access           | W6         | Two-dimensional P/C model, 10 rights, PO, controlled view |
| pkg/ai/gateway                  | W9.3, W9.4 | AI Evidence Gateway + contribution record                 |

**192 new tests.** Build, vet and gofmt clean.

## 2. The anti-duplication audit (MIP §7)

§7 is the rule most likely to be violated by an agent asked to implement a large plan: build the thing named, whether or not it exists. Before writing any code, each capability the MIP names was checked against the repository.

## Found present and REUSED — no fork, no second engine:

| Capability                 | Existing implementation                             |
|----------------------------|-----------------------------------------------------|
| Event ledger               | pkg/platform/audit (hash-chained, Merkle, verified) |
| Policy plane               | pkg/authz                                           |
| Trust calculus             | pkg/core/trustcalc, pkg/trust                       |
| Contradiction              | pkg/insurance/contradiction                         |
| Hypothesis                 | pkg/insurance/causation                             |
| Dependency / corroboration | pkg/insurance/evidence/*.go                         |
| Retention / legal hold     | pkg/governance/data                                 |
| Replay                     | pkg/replay, Store.Replay                            |
| Verifier                   | pkg/commercial/packageverify                        |
| Canonicalization           | pkg/canonical/jcs                                   |
| Evidence manifest FSM      | pkg/evidence/manifest (FROZEN)                      |

**Forbidden engines from §7 — none created:** AuditEngine, DissentEngine, NeutralityEngine, ProtectiveOrderEngine, ProvenanceEngine, DisclosureEngine, RedactionEngine.

**No second ledger and no second policy engine were added.** The event package defines a canonical envelope and hash discipline; the authoritative store remains pkg/platform/audit. The disclosure and AI packages compose pkg/authz's purpose binding rather than replacing it.

## 3. Design decisions worth recording

Three choices where the obvious implementation would have been wrong.

### NOT\_EVALUABLE is the zero verdict

pkg/constitution has three verdicts, and the third is load-bearing. NOT\_EVALUABLE means the facts supplied do not carry what the article needs — and it is **never counted as a pass**. It is the zero value, so a check that forgets to set a verdict reports "cannot judge", never "passes".

This is the constitutional analogue of the independent verifier's SKIP, and it has a cost: an empty subject violates nothing while leaving most articles unjudged. NoViolations() therefore deliberately does **not** assert compliance, and a test exists specifically to fail if that ever changes.

### UNKNOWN is not INDEPENDENT — and it is expensive

Article 28 means an unassessed dependency dimension leaves a source pair UNKNOWN, which cannot satisfy a two-source requirement. A great many plausible corroboration claims degrade to UNKNOWN under this rule. That is the correct and honest outcome; a system resolving unknowns optimistically reports corroboration it does not have.

Cluster() follows the same discipline in both directions: only a *positively established* dependency merges sources. An UNKNOWN pair is reported, not silently decided either way.

### Hard filters before optimization

nextbest removes a rights-denied candidate from the set rather than down-weighting it. If rights were a term in the priority ratio, a sufficiently valuable piece of evidence would eventually outrank its own illegality. Article 4 is not a coefficient.

The test for this uses a candidate with maximal diagnostic value and near-zero cost, and proves it still loses to a modest lawful one.

### The vocabulary has no PROVEN

pkg/qualification/state deliberately contains no `PROVEN`, `ESTABLISHED`, `CORROBORATED` or `LIABLE`. That absence is the enforcement mechanism for Article 16 — a vocabulary that cannot express a verdict cannot accidentally deliver one, and no downstream editorial care is needed to keep the boundary. The forbidden terms are held as data so the prohibition is testable, and `Parse` refuses them with a distinct error from "unknown".

## 4. FILES

```
docs/architecture/VERIQO_OS_ARCHITECTURE.md
docs/constitution/EVIDENCE_CONSTITUTION.md
docs/ieap/IEAP-001.md
docs/eqf/EQF-001.md
docs/nep/NEP-001.md
docs/ai/AI-AUTHORITY-001.md
docs/mip/MIP-001.md

pkg/constitution/{constitution.go, checks.go, constitution_test.go}
pkg/contract/event/{event.go, event_test.go}
pkg/qualification/observability/{observability.go, observability_test.go}
pkg/qualification/independence/{independence.go, independence_test.go}
pkg/qualification/state/{state.go, state_test.go}
pkg/qualification/reverseproof/{reverseproof.go, reverseproof_test.go}
pkg/qualification/nextbest/{nextbest.go, nextbest_test.go}
pkg/disclosure/access/{access.go, access_test.go}
pkg/ai/gateway/{gateway.go, gateway_test.go}
test/adversarial/constitutional_adversarial_test.go
```

## 5. CONTRACTS

```
constitution.Article/Subject/Result · event.Envelope/Chain · observability.Assessment/Result/State ·
independence.Source/Assessment/Verdict · state.State/SingleSourceException/Qualification ·
reverseproof.Claim/Requirement/RequirementSet/Gap · nextbest.Candidate/Ranking ·
access.Grant/Request/Decision/ControlledViewSession/ProtectiveOrder ·
gateway.Policy/Request/Decision/Projection/Contribution
```

## 6. POLICIES

One policy plane, extended — not replaced. `access.Grant` carries the disclosure lattice; `gateway.Policy` carries model allowlists, jurisdictions, AI-permitting licences and purpose → field maps. Both compose `pkg/authz`.

## 7. EVENTS

25 canonical families (§11) validated by `event.Validate`. Actor types `HUMAN` / `SERVICE` / `AI` / `SYSTEM` — AI is separately named so an AI-caused event can never be indistinguishable from a human one.

## 8. STATE MACHINES

Absence states (7, one weight-bearing) · Independence verdicts (3) · Qualification states (10) · Privilege lifecycle (9) · Requirement status (4) · Procedural P0–P5 · Content C0–C5.

## 9. TESTS

| Package            | Tests |
|--------------------|-------|
| pkg/constitution   | 43    |
| pkg/contract/event | 21    |

| Package               | Tests |
|-----------------------|-------|
| pkg/qualification/*   | 72    |
| pkg/disclosure/access | 23    |
| pkg/ai/gateway        | 20    |
| test/adversarial      | 13    |
| Total new             | 192   |

## 10. RACE TEST

`go test -race` passes across the new packages and the full suite.

## 11. ADVERSARIAL

`test/adversarial` covers, from §23: shared upstream source (all three routes), AI direct connector attempt, AI trust/policy manipulation, privilege leakage, asymmetric access, party credential attack, source steering, dissent suppression, policy retroactivity, ledger tampering (both directions), hash mismatch, raw artifact missing, parsed response surviving without raw, source unavailable, reviewer conflict, commercial influence, cross-tenant access.

Per-package suites additionally cover redaction recovery, OCR/metadata recovery framing, acknowledgement manipulation, and narrative injection.

## 12. REPLAY

The existing replay fabric is reused unchanged. `event.VerifyChain` is a pure function over a slice, so an independent verifier can run it on an exported log with no VERIQO runtime — Article 10.

## 13. SECURITY

Fail-closed throughout. Notable: an empty model allowlist approves nobody rather than everybody; a licence must *positively* permit AI processing; privileged material default-denies including to AI; and `Chain.Append` assigns sequence and predecessor rather than trusting the caller, so a caller cannot fork the chain.

## 14. KNOWN LIMITATIONS

Stated plainly, per §32 rule 18.

| Limitation                                              | Detail                                                                                                                                                                                           |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The new fabric is not yet wired into the Commercial API | The nine packages are real, tested and composable, but no <code>/v1/...</code> route yet drives them end to end. They are libraries with proven semantics, not a running pipeline.               |
| Aureum AI and God of EYS do not exist                   | The gateway that constrains them is built and enforced. The intelligences are not.                                                                                                               |
| No dark web acquisition                                 | Blocked on jurisdiction-specific legal review of acquisition authority. Building it without that review would itself violate Article 4.                                                          |
| No ZKP prover                                           | Article 9's bounded-meaning rule is expressed; no prover ships, and none is claimed.                                                                                                             |
| No redaction media workers                              | The state machine and assurance record are specified. PDF/XLSX/PPTX workers and an adversarial recovery lab are not built, so Article 18 reports <code>NOT_EVALUABLE</code> rather than passing. |
| Case Constitution object not built                      | Policy pinning primitives exist; the bound, signed Case Constitution is designed only.                                                                                                           |
| Conflict registry is a check, not a service             | Article 14 is checkable; no standing registry exists.                                                                                                                                            |

| Limitation                                | Detail                                                                                             |
|-------------------------------------------|----------------------------------------------------------------------------------------------------|
| Common Fact Pack is a subset              | Evidence Dossier v1 covers part of the 25-section specification.                                   |
| IEAP rights-before-contact is unexercised | The rule is constitutional and checked, but no live connector performs a real acquisition to gate. |

## 15. PRODUCTION BLOCKERS

Unchanged, all eight external, all `READY_FOR_REAL_QUALIFICATION` — which is **not** `QUALIFIED`:

|                        |                                       |
|------------------------|---------------------------------------|
| B1 HSM/KMS tenancy     | B5 100-node physical qualification    |
| B2 Live data contract  | B6 72-hour soak                       |
| B3 Multi-region        | B7 SPIFFE/mTLS production attestation |
| B4 Independent pentest | B8 Live vulnerability feed            |

## 16. CLAIMS THAT MUST NOT BE MADE

Per §38, stated explicitly so no downstream document overstates this round:

- **Not** `PRODUCTION_QUALIFIED`, and not "production ready".
- **Not** "the Evidence OS is complete" — W2, W7, W10, W12 are partial or designed.
- **Not** "AI-governed evidence" — the gateway constrains AI; by construction no AI holds evidence authority.
- **Not** "dark web integrated", "ZKP capability shipped", or "redaction verified".
- **Not** "independent corroboration" for any same-root or `UNKNOWN` pair.
- **Not** "constitutionally compliant" from an absence of violations — `NOT_EVALUABLE` articles are unjudged, not passed.
- **Not** "MLETR-qualified" — a legal opinion this engagement cannot produce.

## 17. Status per §34

|                                    |                                                                   |
|------------------------------------|-------------------------------------------------------------------|
| Architecture freeze (W0) .....     | IMPLEMENTED                                                       |
| Constitution (executable) .....    | IMPLEMENTED · UNIT_TESTED · ADVERSARIAL_TESTED                    |
| Event envelope .....               | IMPLEMENTED · UNIT_TESTED                                         |
| EQF (5 components) .....           | IMPLEMENTED · UNIT_TESTED · ADVERSARIAL_TESTED                    |
| Disclosure P/C + PO + view .....   | IMPLEMENTED · UNIT_TESTED · ADVERSARIAL_TESTED                    |
| AI evidence gateway .....          | IMPLEMENTED · UNIT_TESTED · ADVERSARIAL_TESTED                    |
| Canonical evidence state .....     | IMPLEMENTED · ADVERSARIAL_TESTED · REPLAY_VERIFIED (pre-existing) |
| Ledger .....                       | IMPLEMENTED · ADVERSARIAL_TESTED · REPLAY_VERIFIED (pre-existing) |
| Independent verifier .....         | IMPLEMENTED · INTEGRATION_TESTED (pre-existing)                   |
| IEAP acquisition .....             | DESIGNED (contracts IMPLEMENTED, sources SIMULATED)               |
| Redaction assurance .....          | DESIGNED                                                          |
| Common Fact Pack .....             | DESIGNED (Dossier v1 a subset)                                    |
| Aureum / God of EYS / dark web ... | DESIGNED                                                          |
| ZKP .....                          | DESIGNED                                                          |
| Everything external .....          | <code>READY_FOR_REAL_QUALIFICATION</code>                         |

Nothing is `EXTERNALLY_VALIDATED`. Nothing is `PRODUCTION_QUALIFIED`.

## 18. Closing

Doc 1 asked for a coherent architecture rather than a Frankenstein platform, and doc 2 asked for that architecture's constitutional layer built into the repository without duplicating what was already there.

Both were done in that order. The constitution is executable rather than decorative; the epistemic fabric that was entirely absent now exists and is adversarially tested; the AI boundary is enforced before the intelligences it constrains were built; and no forbidden engine was created — every new capability composes an existing primitive.

What remains is named in §14 and §16, at the level of specificity that lets someone else pick it up: not "more work needed", but which wave, which package, and what external dependency stands in the way.